SESCO CEMENT CORPORATION

Fly Ash
Comprehensive Market Dossier

Coal Combustion Products & Fly Ash as SCM
US Supply/Demand, Import Economics & Trade Flow Analysis

CONFIDENTIAL — INTERNAL USE

Prepared for:

Mr. Rick Van Eyk

Chief Executive Officer

SESCO Cement Corporation

Version 3.0 | February 18, 2026

Prepared by OceanDatum Market Intelligence

Part of the SCM Commodity Dossier Series

Contents

1. Types of Fly Ash
2. US Production & Domestic Supply
3. The Coal Plant Closure Crisis
4. Global Production & Market Size
5. Imports, Exports & Trade Flows
6. Pricing
7. SCM Market Context
8. Major Producers & Suppliers
9. End Uses by Application
10. Physical Properties & Specifications
11. Carbon & Sustainability
12. Regulatory & Standards
13. Logistics & Handling
14. Industry Trends & Outlook
15. Quick Reference — Key Numbers

FLY ASH COMPREHENSIVE DOSSIER

Prepared: February 18, 2026 | **Version:** 3.0 | **Focus:** Coal Combustion Products & Fly Ash — US Supply/Demand, Import Economics & Trade Flow Analysis

EXECUTIVE SUMMARY: Fly ash is the most widely used supplementary cementitious material (SCM) in US concrete, but faces a structural supply crisis as coal-fired power generation declines. US coal plant capacity has fallen 48% from the 2011 peak (318 GW) to ~165 GW (end 2025), with an additional 68.8 GW scheduled to retire by 2030. Total CCP production fell from 111 Mt (2017) to 67 Mt (2023). Beneficial use reached a record 72% (2024) but total available fly ash for concrete is shrinking. US prices have risen to $118–123/Mt while Asian FOB prices remain $16–27/Mt, creating a $91–107/Mt import arbitrage. South Korea (49%), Turkey (26%), and China (23%) are the primary import sources. The fly ash supply gap is projected to widen to 12–18 Mt by 2028–2030, driving increased reliance on imports, reclaimed ash from landfills, and alternative SCMs.

67 Mt
Total CCP Produced (2023)
-16% from 80 Mt (2022)

72%
Beneficial Use Rate (2024)
Record — 10th yr >50%

$123/Mt
US Fly Ash Price (2025)
+$5 YoY

165 GW
US Coal Capacity (End 2025)
-48% from 2011 peak

Executive Briefing — Strategic Implications for SESCO Cement

The fly ash market presents both challenges and opportunities for SESCO's cement import and distribution operations. This briefing highlights the five most critical takeaways:

- **Domestic fly ash supply is in structural decline.** US coal capacity has fallen 48% since 2011 and will lose another 68.8 GW by 2030. Total CCP production dropped from 111 Mt (2017) to 67 Mt (2023). This is permanent — coal plant closures are irreversible. The SCM most widely used in US concrete is disappearing.
- **Import arbitrage is compelling.** US fly ash trades at $118–123/Mt while Asian FOB prices are $16–27/Mt. Even with ocean freight ($30–50/Mt), landed cost of $46–77/Mt represents 37–61% savings vs domestic. **SESCO opportunity:** SESCO's Houston terminal infrastructure and vessel agency expertise position the company for fly ash import/distribution.
- **Tariff barriers are selective.** Chinese fly ash faces a 145% tariff, effectively blocking that origin. South Korean (49% of imports) and Turkish (26%) fly ash face minimal barriers. **SESCO opportunity:** Turkey is already a SESCO cement supply partner (Nuh Cement); fly ash co-sourcing from Turkey is a natural extension.
- **Reclaimed ash is a growing domestic source.** Previously disposed ash reclamation has doubled from 2.5 Mt (2021) to 5.0+ Mt (2024 est). Companies like Eco Material Technologies (EMT) and Charah Solutions are commercializing landfill ash processing. Quality variability and ASTM compliance remain challenges.
- **The combined SCM deficit is accelerating.** Fly ash supply is declining (coal closures) while GGBFS supply is declining (EAF transition). US cement demand continues rising (IIJA $1.2T through 2028). This dual SCM crisis creates sustained pricing pressure and import demand. **SESCO opportunity:** Building a diversified SCM import portfolio (fly ash + GGBFS + natural pozzolans) alongside cement creates a differentiated competitive position.

1. TYPES OF FLY ASH

Type	Source	Key Properties	Primary Uses	SCM Value
Class F	Anthracite or bituminous coal combustion	Pozzolanic; $SiO_2+Al_2O_3+Fe_2O_3$ ≥70%; CaO <18%; low LOI	General concrete SCM (15–35% replacement); mass concrete; marine structures; sulfate resistance	**HIGHEST — Premium SCM**
Class C	Lignite or sub-bituminous coal combustion	Pozzolanic + cementitious; $SiO_2+Al_2O_3+Fe_2O_3$ ≥50%; CaO 18–40%; self-cementing	Concrete SCM (15–30% replacement); soil stabilization; flowable fill	**HIGH — Self-cementing advantage**

Class N (Natural Pozzolan)	ASTM C618 also classifies raw/calcined natural pozzolans	$SiO_2+Al_2O_3+Fe_2O_3$ ≥70%; not coal-derived; variable	Concrete SCM; cement replacement where fly ash unavailable	**MODERATE — Growing alternative**
Bottom Ash	Coarser fraction from coal combustion; falls to boiler bottom	Coarse, granular; 10–20% of total CCP; angular particles	Structural fill; aggregate substitute; roadbase	LOW — Aggregate only
Boiler Slag	Wet-bottom boilers; molten ash quenched in water	Glassy, angular; <2% of CCP; declining as wet-bottom boilers retire	Blasting abrasive; roofing granules; fill	LOW — Niche
FGD Gypsum	Flue gas desulfurization scrubbers	$CaSO_4 \cdot 2H_2O$; high purity; ~15% of CCP volume	Wallboard manufacturing; cement retarder; agriculture	LOW — Different market (gypsum)

CRITICAL DISTINCTION: Only Class F and Class C fly ash have significant SCM value for concrete. Class F (from bituminous coal, dominant in eastern US) is the premium product for durability applications. Class C (from sub-bituminous coal, dominant in western US) has self-cementing properties but higher CaO limits its use in some applications. The ~14.5 GW retiring in 2025 is primarily MISO and PJM capacity (Midwest/Mid-Atlantic) — the heartland of Class F production.

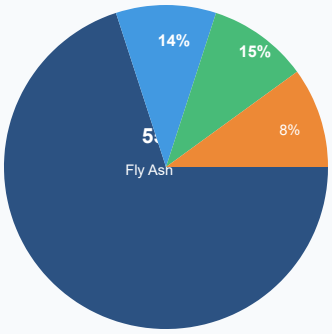
2. US PRODUCTION & DOMESTIC SUPPLY

2.1 Coal Combustion Products Production

Year	CCP Production (Mt)	Beneficial Use (Mt)	Use Rate	Disposed (Mt)
2015	107	56	52%	51
2017	111	71	64%	40
2019	79	41	52%	38
2020	69	41	59%	28
2021	77	35	46%	42
2022	80	50	62%	30
2023	67	46	69%	21
2024 (est)	~60	~43	72%	~17

2.2 Fly Ash Breakdown within CCP

US CCP Production by Type (2023 — 67 Mt Total)



- Fly Ash — 36.9 Mt (55%)
- Bottom Ash — 9.4 Mt (14%)
- FGD Gypsum — 10.1 Mt (15%)
- FGD Other — 5.4 Mt (8%)
- Boiler Slag/Other — 5.2 Mt (8%)

KEY: Fly ash = 55% of CCP by weight
~37 Mt fly ash produced from 67 Mt total CCP (2023)

Source: ACAA 2023 Production & Use Survey. Fly ash is the single largest CCP category. Concrete/grout consumes 61% of beneficially used fly ash.

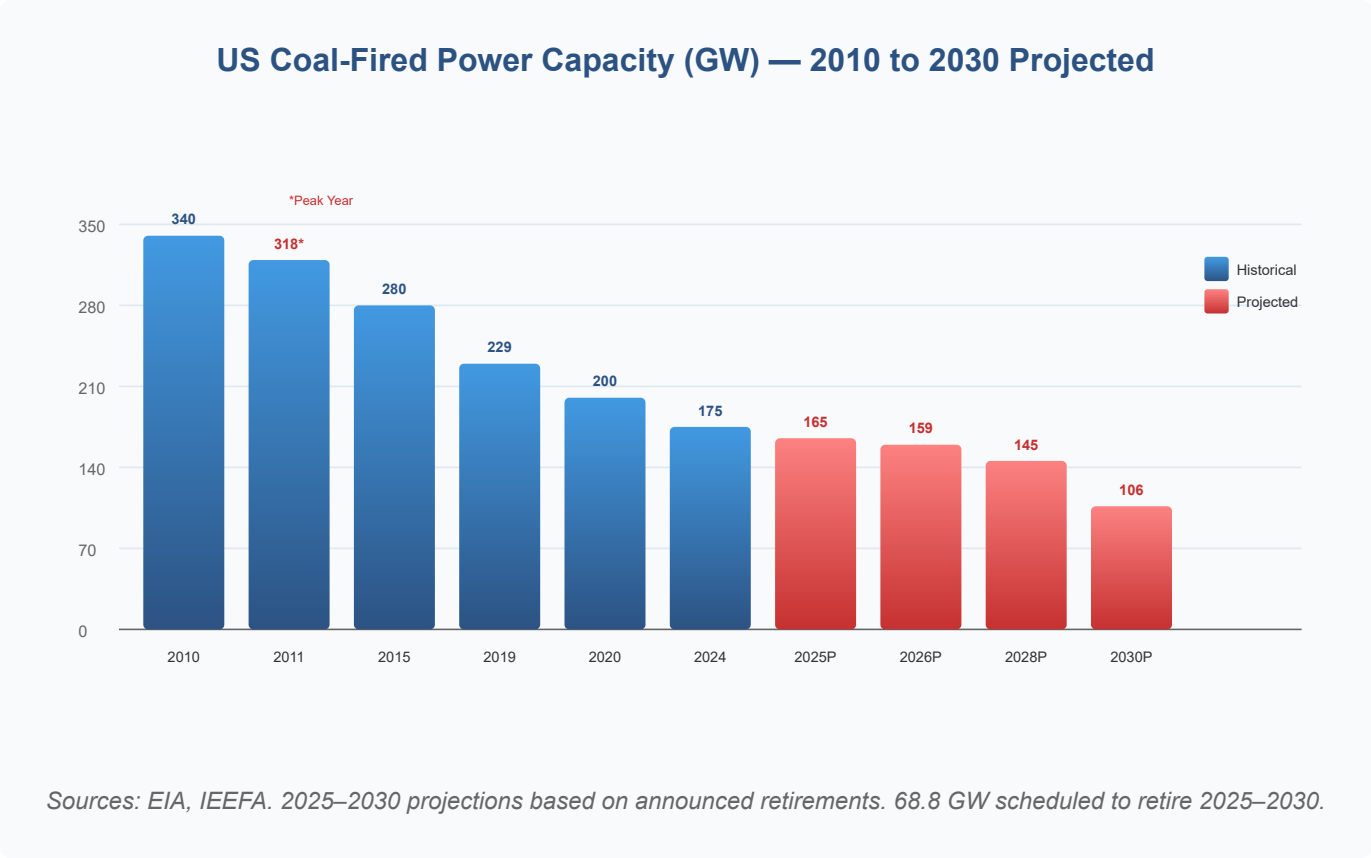
2.3 Top Fly Ash Producing States

State	Coal Capacity (GW, 2024)	Fly Ash Class	Notes
Texas	~18	Mixed F/C	Largest coal fleet; multiple units retiring 2025–2028
Indiana	~14	Class F (bituminous)	Heavy coal reliance; MISO region; significant retirements
West Virginia	~12	Class F (bituminous)	Appalachian coal heartland; premium Class F source
Ohio	~10	Class F (bituminous)	PJM region; Duke Beckjord ash reclamation opportunity
Kentucky	~9	Class F (bituminous)	TVA and utility plants; major retirement wave
Wyoming	~7	Class C (sub-bituminous)	PRB coal region; Class C source for western US
Missouri	~7	Mixed F/C	Ameren, Evergy fleets
Georgia	~6	Class F	Southern Co. fleet; Plant Scherer, one of the largest

Pennsylvania	~6	Class F (anthracite/bituminous)	PJM region; premium Class F
Alabama	~5	Class F	TVA and Alabama Power

GEOGRAPHIC NOTE: Class F fly ash (from bituminous/anthracite coal) is concentrated in eastern states: WV, OH, PA, IN, KY, GA, AL. Class C fly ash (from sub-bituminous coal) is concentrated in western states: WY, MT, ND, TX. The 2025 retirement surge hits MISO (39%) and PJM (27%) hardest — primarily Class F production territory.

3. THE COAL PLANT CLOSURE CRISIS



3.1 2025 Retirement Surge

Metric	2023	2024	2025 (Projected)	Change
Capacity Retired (MW)	10,577	3,629	14,532	+300% YoY
Coal Consumed by Retiring Units (M short tons)	28.5	9.0	39.6	+340% YoY
Primary Regions Affected	MISO (39%), PJM (27%), SPP (15%)			

2025 RETIREMENT SURGE: Record 14.5 GW retiring in 2025 — the highest annual retirement since the coal decline began. These units consumed 39.6 Mt of coal in 2023, generating millions of tons of fly ash. MISO + PJM account for 66% of 2025 retirements, removing significant Class F fly ash production from the Midwest and Mid-Atlantic — precisely the regions where concrete demand remains strong.

3.2 Supply Impact Summary

Supply Threats

- **Coal retirements:** 68.8 GW scheduled to close 2025–2030
- **Production decline:** 111 Mt CCP (2017) → 67 Mt (2023) → ~45 Mt by 2030
- **Capacity factor:** Dropped from 67% (2010) to 42% (2023)
- **Regional loss:** MISO + PJM losing 66% of 2025 retirements (Class F heartland)
- **EPA regulations:** Mercury/Air Toxics (MATS), CCR Rule driving closures

Supply Stabilizers

- **Beneficial use rate:** 72% (2024 record) — extracting more from less
- **Reclaimed ash:** 4.5 Mt (2023), doubling annually
- **Imports growing:** Korea, Turkey filling domestic gap
- **CCS at coal plants:** Could extend some plant lifespans
- **Western coal:** PRB plants retiring more slowly (Class C)

4. GLOBAL PRODUCTION & MARKET SIZE

4.1 World Fly Ash Production

Region	CCP/Fly Ash Production (Mt/yr)	Utilization Rate	Key Detail
China	~600 Mt fly ash	~68%	World's largest producer; 145% tariff on US imports
India	~280 Mt CCP	~80%	Rapidly growing utilization; government mandates
EU-27	~40 Mt	>90%	Mature market; coal phase-out accelerating (Germany, Poland last major)
United States	~67 Mt CCP (37 Mt fly ash)	72%	Structural decline; coal closures irreversible
Japan/S. Korea	~20 Mt combined	>95%	Export surplus; high quality; key US import sources
Australia	~13 Mt	~45%	Lower utilization; export potential to Asian markets

SE Asia	~50 Mt	~30–40%	Vietnam, Indonesia, Philippines — low-cost FOB source

4.2 Global Market Size

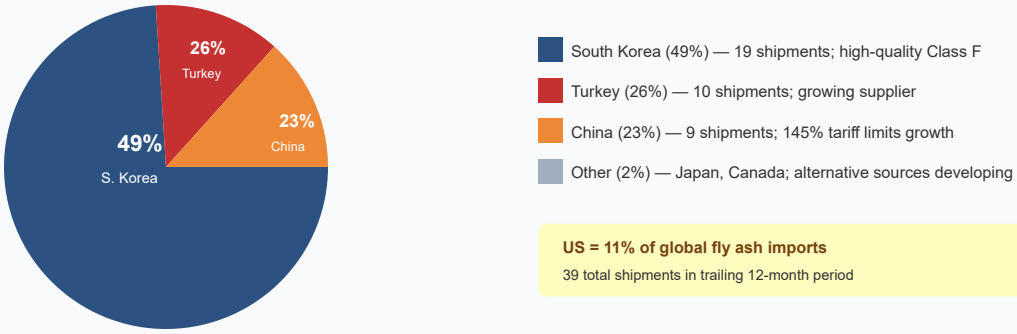
Market Segment	2024 Value	Forecast	CAGR
Global Fly Ash Market	$5.2–7.4 billion	$8.7–12.5B by 2031–33	5.2–7.5%
US Fly Ash Market	$2.36 billion	$3.2B by 2030	5.2%
Global SCM Market	$91.1 billion	$119.9B by 2033	3.1%

MARKET PARADOX: The US fly ash market value continues to grow ($2.36B, +5.2% CAGR) even as production volume declines. This reflects rising prices driven by supply scarcity. China produces ~600 Mt of fly ash annually (16x the US) but faces 145% tariff barriers. South Korea and Japan, with utilization rates >95%, have limited export surplus but are the primary suppliers to the US.

5. IMPORTS, EXPORTS & TRADE FLOWS

5.1 US Import Position

US Fly Ash Imports by Source Country (Nov 2023 – Oct 2024)



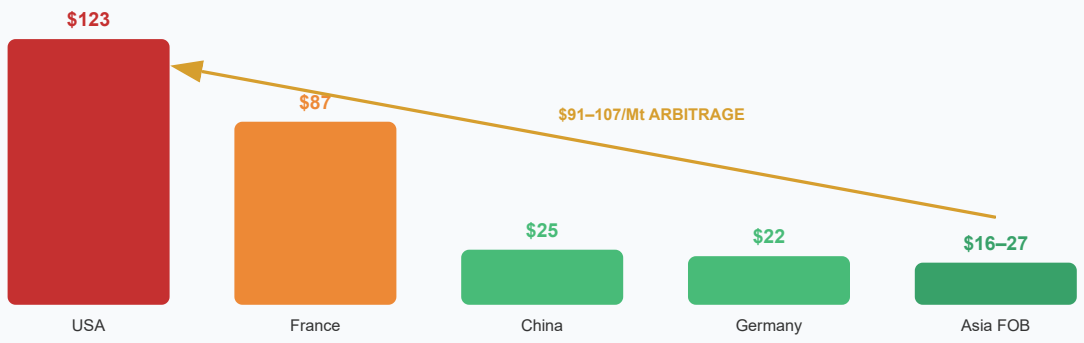
Source: Volza Import Statistics (Nov 2023 – Oct 2024). US accounts for 11% of global fly ash import volume.

5.2 Key Receiving States

State	Role	Drivers
Florida	Major importer	No domestic coal plants; massive construction; port access
Washington	West Coast gateway	Trans-Pacific route from Korea/Japan; short sea from BC
Louisiana	Gulf gateway	Mississippi River system distribution; lower landing costs
Texas	Gulf gateway + domestic	Houston terminal infrastructure; local coal declining
Nevada	Inland importer	No domestic coal; rail from West Coast ports
Ohio	Domestic + reclaimed	Beckjord reclamation; Midwest concrete demand

5.3 Import Economics

Fly Ash Price Comparison by Region ($/Mt — 2025)



Sources: ChemAnalyst, IMARC Group (2025). Asian FOB + ocean freight ($30–50/Mt) = landed cost $46–77/Mt vs US domestic $118–123/Mt. Import savings of 37–61%.

TARIFF LANDSCAPE: Chinese fly ash faces **145% tariff** (Sec. 301), effectively blocking that origin despite $16–25/Mt FOB pricing. South Korean and Turkish fly ash face standard duties only (minimal). Japan and Canadian fly ash enter at favorable rates. The tariff structure creates a two-tier import market: Korea/Turkey accessible, China blocked.

6. PRICING

6.1 Current Pricing (2025)

$123/Mt
USA (Q3 2025)
+$5 YoY

$118/Mt
USA (Q2 2025)
-5.3% QoQ

$87/Mt
France
Stable

$22/Mt
Germany
Oversupply

$16–27/Mt
Asian FOB
Abundant supply

6.2 Historical US Pricing Trend

Period	US Price ($/Mt)	Key Driver
2018	~$50–70	Adequate supply; coal fleet still ~230 GW
2020	~$70–85	Coal capacity dropped below 200 GW; tightening supply
2022	~$90–100	Post-COVID construction surge; supply-demand gap widening
2024 Q2	$118	Coal retirements accelerating; IIJA spending ramping
2025 Q3	$123	Record retirements (14.5 GW); constrained supply + strong demand

6.3 Price Drivers

Factor	Impact	Direction
Coal plant closures	Reduced domestic supply	Price Up
IIJA infrastructure spending	Sustained concrete demand	Price Up
145% China tariff	Reduced cheapest import source	Price Up
Buy Clean / LEED v5 mandates	Increased SCM demand	Price Up
Declining Asia-USWC freight	Lower import costs	Price Down
Korean/Turkish import growth	Increased competition	Price Down

Reclaimed ash expansion	New domestic supply source	Price Down

7. SCM MARKET CONTEXT

7.1 Fly Ash vs Competing SCMs

SCM	US Consumption	Typical Replacement	Supply Trend	Price ($/Mt)
Fly Ash	~15–35 Mt	15–35%	Declining (coal closures)	$118–123
GGBFS	~3–5 Mt	25–70%	Declining (BF→EAF shift)	$100–140+
Natural Pozzolans	Growing	15–25%	Growing (volcanic, pumice)	$40–80
Silica Fume	<1 Mt	5–10%	Stable (niche)	$400–600+
LC3 (Calcined Clay)	Minimal	Up to 40%	Emerging (early US stage)	$50–80 est.
Portland Cement	~100 Mt	Baseline	Stable/growing	$163

DUAL SCM SUPPLY CRISIS: The US faces a unique crisis — both primary SCMs are in structural decline simultaneously from unrelated industrial transitions. Fly ash is declining because coal plants are closing. GGBFS is declining because blast furnaces are being replaced by EAFs. Combined, this creates a growing SCM deficit that cannot be filled by domestic sources alone. LC3 (limestone calcined clay cement) is the leading long-term replacement, but US commercial-scale adoption remains minimal.

7.2 Fly Ash Performance in Concrete

Advantages

- 15–35% cement replacement (typical); higher in mass concrete
- Improved workability and pumpability
- Reduced water demand (ball-bearing effect of spherical particles)
- Reduced heat of hydration (mass concrete, hot weather)
- Enhanced sulfate resistance (Class F)
- Reduced ASR (alkali-silica reaction) expansion
- Lower permeability at later ages

Limitations

- Slower early strength gain (3 and 7-day)
- Quality variability by coal source and plant
- LOI (loss on ignition) can vary — high LOI reduces air entrainment
- Supply increasingly constrained and unreliable
- Class C high-CaO limits sulfate resistance
- Cold weather extends set times further
- Color variability (aesthetic applications)
- Fineness varies between sources

- 86–95% lower embodied CO2 than OPC
- LEED v5 / Buy Clean compliance
- Cost savings vs Portland cement ($40–60/Mt less)

---

8. MAJOR PRODUCERS & SUPPLIERS

8.1 Ash Processors & Marketers

Company	Role	Key Details
Eco Material Technologies (EMT)	Largest US ash processor	Boral Industries ash business acquired. 12+ Mt/yr handling. Reclaimed ash processing (harvesting). Proprietary beneficiation technology (LOI reduction, classification). 40+ terminals/processing sites.
Charah Solutions	Ash management & processing	NYSE: CHRA. Environmental services + ash marketing. MultiCem™ slag/pozzolan product. Greens Port terminal (Houston). Coal ash pond remediation.
SEFA Group	Ash marketer	Southeast/Mid-Atlantic focus. Long-term utility contracts. Duke Energy, Dominion supply agreements.
Headwaters (now Boral)	Former major processor	Acquired by Boral 2017; ash business to EMT. Legacy contracts at many plants.
Salt River Materials Group	Southwest processor	Arizona/New Mexico focus. Navajo Generating Station ash inventories.
ISG Resources	Midwest/South processor	Utility-owned ash marketing. Quality control focus.

8.2 Major Utility Producers (Coal Fleet Owners)

Utility	Coal Fleet (GW, approx)	Key Plants	Retirement Plans
Duke Energy	~8	Belews Creek, Marshall, Gibson	Aggressive coal exit; Beckjord closed 2014 (reclamation site)
Southern Company	~10	Plant Scherer (3.5 GW), Miller, Bowen	Scherer one of largest US coal plants; partial unit retirements
AEP (American Electric Power)	~12	Mountaineer, Mitchell, Cardinal	Exiting coal by 2035; WV plants among last

Ameren	~5	Labadie, Rush Island	Rush Island closed; Labadie 2036
TVA (Tennessee Valley Authority)	~6	Cumberland, Kingston, Paradise	Federal mandate; aggressive closure schedule
Vistra Energy	~8	Luminant TX fleet, various Midwest	Texas lignite plants among first to retire

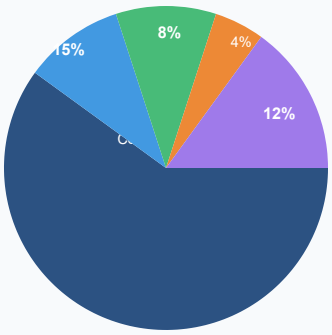
8.3 International Suppliers

Company/Source	Country	Quality	US Market Strategy
KEPCO/Korean generators	South Korea	High-quality Class F	49% of US imports; consistent quality; established logistics
Turkish utilities	Turkey	Class F	26% of US imports; growing rapidly; existing cement trade lanes
Chinese generators	China	Variable	23% of imports but 145% tariff limits growth; cheapest FOB
J-Power/JERA	Japan	Premium Class F	Small share but high quality; potential growth

9. END USES BY APPLICATION

9.1 Fly Ash Beneficial Use by Application (2023)

2023 Fly Ash Beneficial Use by Application (46.3 Mt total CCP beneficially used)



- Concrete/Grout — 28.2 Mt (61%)
- Structural Fills — 6.9 Mt (15%)
- Road Base/Stabilization — 3.7 Mt (8%)
- Mine Backfill — 1.9 Mt (4%)
- Other (gypsum panels, cement kiln, etc.) — 5.6 Mt (12%)

FLY ASH IN CONCRETE +9% YoY
11.9 Mt (2023) vs 10.9 Mt (2022) — demand rising

Source: ACAA 2023 Production & Use Survey. Fly ash in concrete increased +9% YoY despite total CCP declining 16%.

9.2 Concrete Applications by Sector

Sector	Fly Ash Use	Typical Replacement %	Key Requirement
Ready-mix concrete	Largest consumer	15–25%	Workability, cost savings, strength
Precast/prestressed	Growing	15–20%	Surface finish, early strength, consistency
Mass concrete (dams, foundations)	Essential	25–40%	Low heat of hydration
Highway paving	DOT specifications	15–25%	Durability, ASR mitigation, sulfate resistance
Oil well cementing	Specialty	Variable	Temperature resistance, low permeability
Soil stabilization	Class C dominant	N/A (direct)	Self-cementing property

10. PHYSICAL PROPERTIES & SPECIFICATIONS

10.1 ASTM C618 Chemical Requirements

Property	Class F	Class C	Class N
$SiO_2 + Al_2O_3 + Fe_2O_3$ (min %)	**≥70%**	**≥50%**	**≥70%**
SO_3 (max %)	5.0	5.0	4.0
Moisture content (max %)	3.0	3.0	3.0
Loss on Ignition (LOI, max %)	6.0*	6.0	10.0

*Some specifications allow up to 12% LOI with approval. Low-LOI ash (<3%) commands premium pricing. High LOI reduces air entrainment in concrete.

10.2 ASTM C618 Physical Requirements

Property	Requirement	Test Method

Fineness (retained on 45µm / No. 325 sieve)	Max 34%	ASTM C430
Strength Activity Index (7-day)	Min 75% of control	ASTM C311
Strength Activity Index (28-day)	Min 75% of control	ASTM C311
Water requirement	Max 105% of control	ASTM C311
Soundness (autoclave expansion)	Max 0.8%	ASTM C151
Uniformity (density variation)	Max ±5% of average	ASTM C311

10.3 Typical Chemical Composition

Oxide	Class F (Bituminous)	Class C (Sub-bituminous)	Portland Cement
SiO2	40–60%	25–45%	17–25%
Al2O3	15–30%	15–25%	3–8%
Fe2O3	5–20%	5–15%	0.5–6%
CaO	1–12%	18–40%	60–67%
MgO	1–4%	3–8%	0.5–6%
SO3	0.5–2%	1–5%	1.5–4.5%
Na2O + K2O (alkali)	1–4%	1–6%	0.2–1.3%
LOI	1–12%	0.5–3%	0.5–3%

10.4 Physical Characteristics

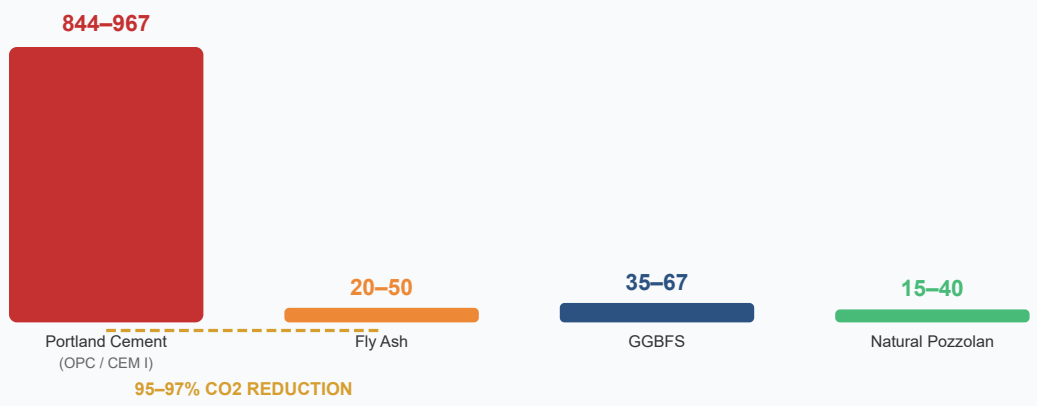
Property	Fly Ash	Portland Cement
Particle shape	Spherical (cenospheres, plerospheres)	Angular, irregular
Particle size	1–100 µm (mean ~20 µm)	1–80 µm (mean ~15 µm)
Specific gravity	2.1–2.6 (Class F); 2.4–2.8 (Class C)	~3.15
Specific surface area	300–500 m²/kg (Blaine)	350–380 m²/kg
Color	Light tan to dark gray (depends on carbon content)	Gray
Hydraulic activity	Pozzolanic (Class F); Pozzolanic + cementitious (Class C)	Direct hydraulic

QUALITY NOTE: The spherical particle shape of fly ash is its key physical advantage — it acts as a ball bearing in concrete, improving workability and reducing water demand. This is unique among SCMs (GGBFS and natural pozzolans have angular particles). Low-LOI Class F fly ash (<3% LOI) from eastern bituminous coal commands premium pricing due to superior air entrainment compatibility.

11. CARBON & SUSTAINABILITY

11.1 Embodied Carbon Comparison

Embodied CO2 by Cementitious Material (kg CO2e / tonne)



Sources: PCA, EPDs, various LCA studies. Fly ash embodied CO2 reflects only collection, processing, and transport energy — combustion emissions allocated to power generation.

11.2 CO2 Reduction in Concrete

Fly Ash Replacement Rate	Concrete CO2 Reduction	Performance Impact
15%	~13–15%	Negligible strength reduction; improved workability
20%	~17–20%	Standard replacement; most DOT specifications
25%	~22–25%	Comparable to OPC at 28 days; improved long-term
35%	~30–33%	Reduced early strength; excellent durability; sulfate resistance
50% (HVFA)	~45–48%	Mass concrete; extended set time; superior durability

CARBON VALUE PROPOSITION: At just 20–50 kg CO2e/tonne, fly ash has the lowest embodied carbon of any major SCM. Every ton of fly ash used instead of Portland cement saves approximately **800–920 kg CO2**. With Buy Clean mandates and LEED v5 now requiring embodied carbon quantification, fly ash is the most impactful ingredient for decarbonizing concrete — but the irony is clear: the cleanest SCM is disappearing because its source (coal combustion) is too dirty.

12. REGULATORY & STANDARDS

12.1 Key Standards

Standard	Scope	Key Provisions
ASTM C618-23	Coal Fly Ash and Natural Pozzolans for Concrete	Defines Class F, Class C, Class N; chemical/physical requirements; SAI testing
ASTM C595/C595M-24	Blended Hydraulic Cements	Type IP (Portland-pozzolan); allows up to 40% pozzolan
ASTM C1157	Performance Hydraulic Cement	Any SCM proportion if performance criteria met
ACI 232.2R	Use of Fly Ash in Concrete	Technical guide: mix design, placement, curing for fly ash concrete
AASHTO M 295	Coal Fly Ash for Highway Concrete	State DOT specification (mirrors C618 with additional state-specific limits)
EN 450-1	Fly Ash for Concrete (European)	European equivalent; Category A/B/C by LOI

12.2 EPA Coal Combustion Residuals (CCR) Rule

Provision	Requirement	Impact on Fly Ash Market
40 CFR Part 257 (2015)	National standards for CCR disposal in landfills and surface impoundments	Increased closure costs incentivize beneficial use
CCR Rule 2024 Amendments	Strengthened groundwater monitoring; legacy pond closure deadlines	Accelerates ash reclamation from legacy ponds
EPA Beneficial Use Determination	Fly ash in concrete classified as **non-waste**; exempt from RCRA	Protects commercial ash market
Comprehensive Procurement Guidelines	Federal agencies must specify fly ash in concrete when available	Demand floor for federal projects

12.3 Buy Clean & Carbon Policy

Regulation	Requirement	Impact on Fly Ash
GSA Low-Embodied Carbon (LEC)	GWP limits for concrete; EPDs required for IRA-funded procurement	**Strong demand driver**
LEED v5 (2024)	Mandatory embodied carbon prerequisite; products must outperform baseline by 20%	**Strong demand driver**
California CALGreen	Maximum GWP for concrete; state-mandated for all public works	**West Coast premium**
State Buy Clean (CO, NY, OR, WA)	EPDs and/or embodied carbon limits for public projects	**Growing demand driver**

REGULATORY TAILWIND: Carbon regulations are simultaneously increasing demand for fly ash while coal regulations are decreasing supply. The IRA's $350B clean energy provisions include Buy Clean mandates for federally funded construction. Since 2023, manufacturers have published **14,000+ additional EPDs for concrete**. Fly ash at 95–97% lower embodied CO2 than OPC is the single most impactful ingredient for meeting these new requirements.

---

13. LOGISTICS & HANDLING

13.1 Transportation Modes

Mode	Application	Details
Ocean Vessel	Import from Korea/Turkey/Asia	Handysize/Supramax (25K–60K DWT); bulk cargo holds; loading ~6,000–10,000 t/day
Barge	Inland waterway distribution	Mississippi system from power plants; ~1,500 tons/barge; covered dry cargo
Rail	Medium-distance distribution	Covered hopper cars; 100–110 ton loads; unit trains from large plants
Truck	Final delivery	Pneumatic tankers (dry bulk); 25–28 ton loads; 100–200 mile typical radius

13.2 Terminal & Storage

Requirement	Specification
Storage type	**Enclosed silos or domes** — moisture is the critical enemy

Silo capacity	Typical 5,000–30,000 tons per silo; import terminals may need 50,000+
Handling system	Pneumatic conveyance (300–500 t/hr); screw conveyors; bucket elevators
Vessel discharge	Grab cranes + hopper/conveyor to silo; 5,000–10,000 t/day rate
Truck loading	Gravity or pneumatic fill into tanker trailers; weighbridge required
Rail loading	Covered hopper car loading via gravity or pneumatic
Quality control	LOI testing, fineness, moisture monitoring at receipt and dispatch
Dust control	Baghouse or cartridge filtration on all transfer points; OSHA PM limits

13.3 Reclaimed Ash Processing

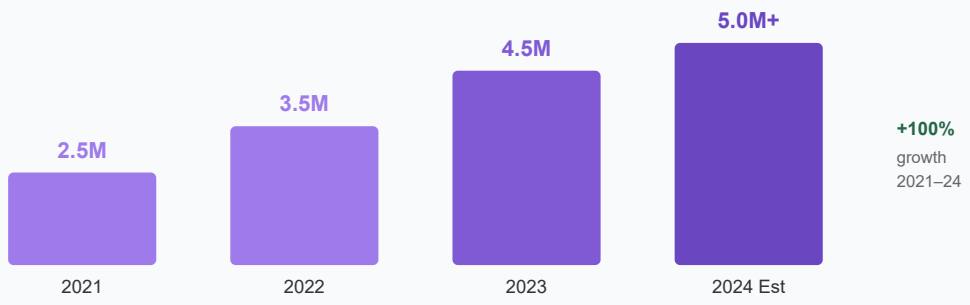
Step	Process	Purpose
1. Excavation	Hydraulic or mechanical dredging from ponds/landfills	Extract raw ash
2. Dewatering	Filter presses, centrifuges, drying beds	Reduce moisture to <3%
3. Classification	Air classification, screening	Remove coarse particles; target <34% retained on 45μm
4. Beneficiation	Carbon burnout or electrostatic LOI reduction	Achieve <6% LOI for ASTM C618 compliance
5. Testing	Full ASTM C618 suite (chemical, physical, SAI)	Certify as Class F or Class C equivalent
6. Storage	Enclosed silo or dome	Moisture-free storage for market

HANDLING NOTE: Fly ash is an extremely fine powder (mean ~20 μm) that becomes airborne readily. All handling must be enclosed with dust collection. Unlike GGBFS (which can be stored open as GBS), fly ash requires fully enclosed pneumatic systems from vessel discharge to truck loading. Terminal infrastructure mirrors cement terminals — silos, pneumatic transfer, covered loading — and can share existing cement import infrastructure. **SESCO's Houston terminal already has the physical infrastructure for fly ash handling.**

14. INDUSTRY TRENDS & OUTLOOK

14.1 Reclaimed Ash — Emerging Domestic Supply

Previously Disposed Ash Reclaimed for Beneficial Use (Million Tons)



Source: ACAA. US has an estimated 1–2 billion tons of fly ash in legacy disposal sites. At current reclamation rates, decades of supply exist — but quality variability and beneficiation costs remain challenges.

14.2 Five-Year Market Outlook (2025–2030)

US Fly Ash Market — 5-Year Outlook

Factor	Trend	Impact on Supply	Impact on Price
Coal Capacity	165 GW (2025) → 106 GW (2030): -36%	NEGATIVE	UP
CCP Production	67 Mt (2023) → ~45 Mt (2030 est)	NEGATIVE	UP
Beneficial Use Rate	72% (2024) → 80%+ (2030 target)	POSITIVE	NEUTRAL
Reclaimed Ash	5 Mt (2024) → 8–10 Mt (2030 est)	POSITIVE	DOWN
Import Volume	Growing — Korea, Turkey, Japan, Canada	POSITIVE	DOWN
IIJA Demand	$62B/year through 2028; concrete +4.5%/yr	N/A	UP
Carbon Regulations	Buy Clean mandates; LEED v5; CALGreen	N/A (demand side)	UP
China Tariff (145%)	Blocks cheapest global source	NEGATIVE	UP

Alternative SCMs (LC3, NP)	Abundant raw materials; early US adoption	**LONG-TERM POSITIVE**	**DOWN (long-term)**

NET ASSESSMENT: The US fly ash market will remain supply-constrained through 2030. Domestic production decline (-33% projected) will outpace demand moderation from beneficial use optimization. The gap will be filled by: (1) increased imports from Korea and Turkey, (2) reclaimed ash from legacy landfills/ponds (5–10 Mt/yr potential), (3) alternative SCMs (GGBFS, natural pozzolans, LC3). Prices will remain elevated in the $100–130/Mt range with periodic spikes during supply disruptions. **The import arbitrage ($91–107/Mt) and terminal infrastructure position create lasting competitive advantages for companies with established bulk handling capabilities and international trade networks.**

15. QUICK REFERENCE — KEY NUMBERS

Metric	Value	Trend
US Price (Q3 2025)	$123/Mt	Up $5 YoY
Asian FOB Price	$16–27/Mt	Stable
Import Arbitrage	$91–107/Mt	Attractive
US Coal Capacity (End 2025)	165 GW	-48% from 2011 peak
Coal Retirements 2025	14.5 GW	Record year
CCP Production (2023)	67 Mt	-16% from 2022
Fly Ash (of CCP, 2023)	~37 Mt (55%)	Declining with coal
Beneficial Use Rate (2024)	72%	Record high (10th yr >50%)
Fly Ash in Concrete (2023)	11.9 Mt	+9% YoY
Reclaimed Ash (2024 est)	5.0 Mt+	+100% from 2021
Legacy Ash Inventory	1–2 billion tons	Decades of potential supply
Top Import Source	South Korea (49%)	Growing
China Tariff	145%	Blocks cheapest source
US Market Size (2024)	$2.36 billion	+5.2% CAGR
Global Fly Ash Market	$5.2–7.4 billion	$8.7–12.5B by 2031–33

IIJA Infrastructure	$1.2T through 2028	Demand driver
Fly Ash Embodied CO2	20–50 kg CO2e/tonne	95–97% lower than OPC
Portland Cement CO2	844–967 kg CO2e/tonne	Industry baseline
CO2 Saved per Ton Fly Ash	~800–920 kg CO2	Key sustainability metric
ASTM C618 Classes	F (bituminous), C (sub-bit), N (natural)	Class F = premium

SOURCES

Industry Organizations: ACAA (American Coal Ash Association) Production & Use Surveys 2015–2024; ARTBA Forecast Update 2024; ACI Committee 232 (Fly Ash in Concrete)
Government: EIA (Energy Information Administration) Coal Capacity Data; USGS CCP Statistics; EPA CCR Rule (40 CFR 257); EPA Comprehensive Procurement Guidelines; DOE Clean Energy Programs
Standards: ASTM C618-23 (Coal Fly Ash and Natural Pozzolans); ASTM C595/C595M-24 (Blended Cements); ASTM C1157 (Performance Cement); ACI 232.2R (Use of Fly Ash in Concrete); AASHTO M 295; EN 450-1
Market Research: ChemAnalyst (Fly Ash Pricing Q2–Q3 2025); IMARC Group (Global Fly Ash Market); Fortune Business Insights; Market Data Forecast; Verified Market Reports; Credence Research
Energy Analysis: IEEFA (Institute for Energy Economics and Financial Analysis) — Coal Capacity Reports 2024–2025; Global Energy Monitor Coal Plant Tracker
Trade Data: Volza Import Statistics (Nov 2023 – Oct 2024); US Census Bureau Foreign Trade; HTS 2621.00
Companies: Eco Material Technologies (EMT); Charah Solutions (NYSE: CHRA); SEFA Group; Salt River Materials; Duke Energy; Southern Company; AEP; TVA; Vistra Energy
Carbon & Sustainability: PCA EPDs; GSA Buy Clean / Low-Embodied Carbon Program; USGBC LEED v5; California CALGreen Code; Climate Earth; various concrete EPDs
Technical: Power Engineering; Concrete Products Magazine; Rock Products; Utility Dive; Engineering News-Record

FLY ASH COMPREHENSIVE DOSSIER v3.0 | February 18, 2026 | Coal Combustion Products & Fly Ash — US Supply/Demand, Import Economics & Trade Flow Analysis
Prepared by OceanDatum Market Intelligence | Part of the SCM Commodity Dossier Series